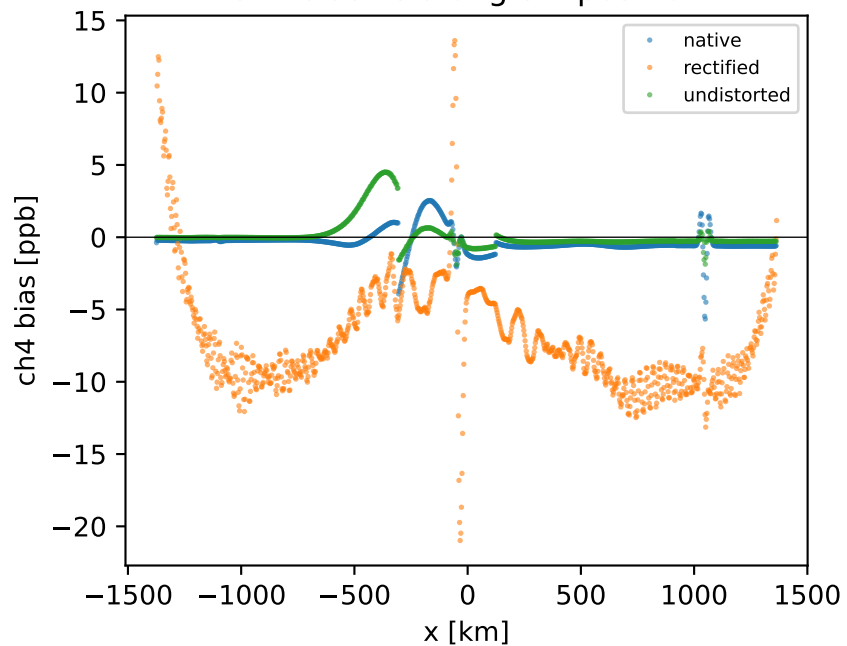
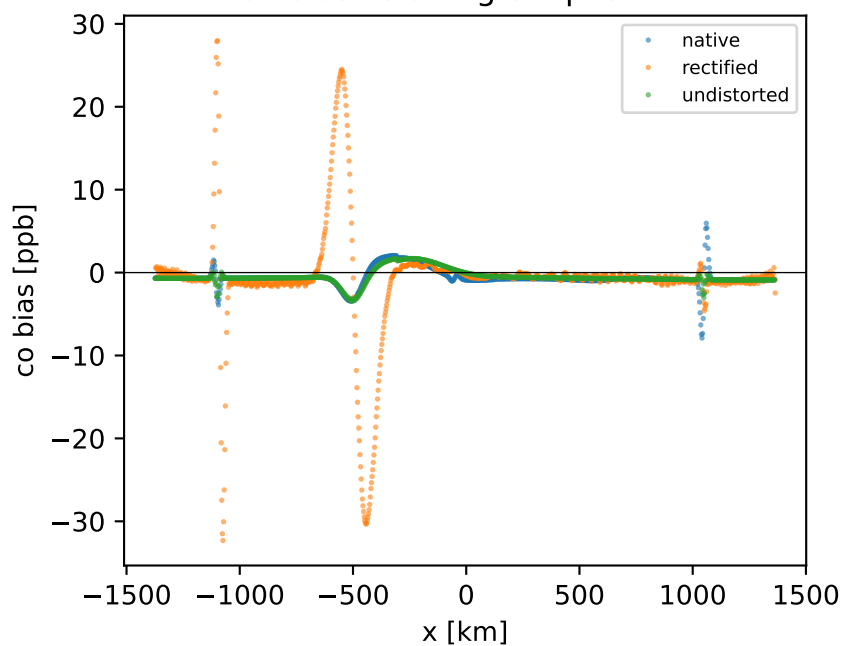


Joint FPA0+FPA3 -- realistic, no noise (native/rectified/undistorted)

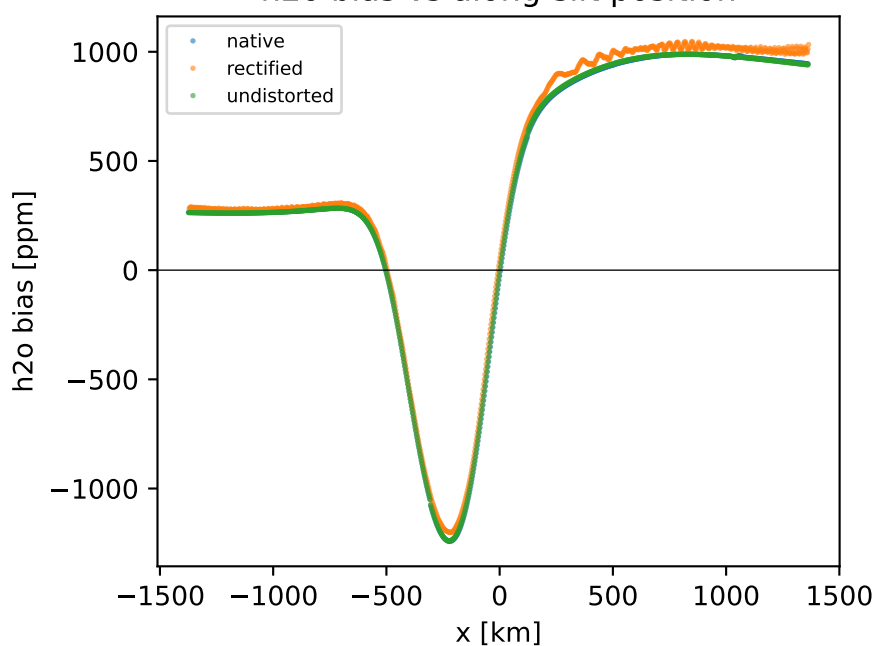
ch4 bias vs along-slit position



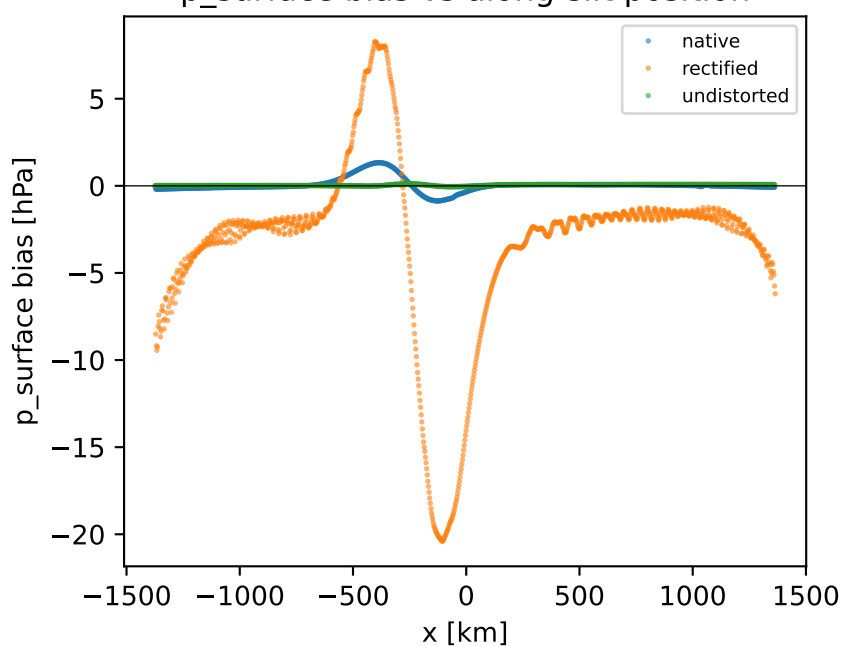
co bias vs along-slit position



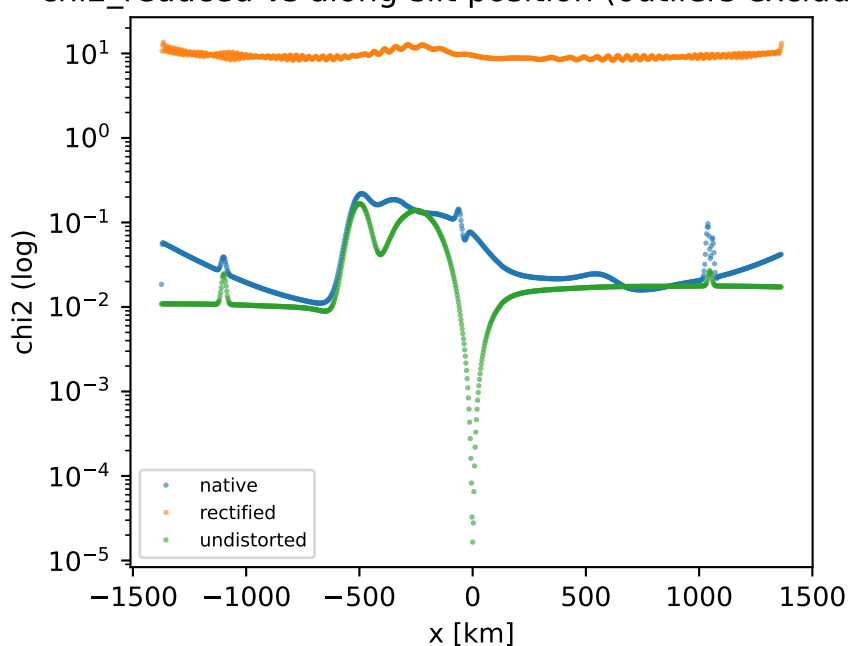
h2o bias vs along-slit position



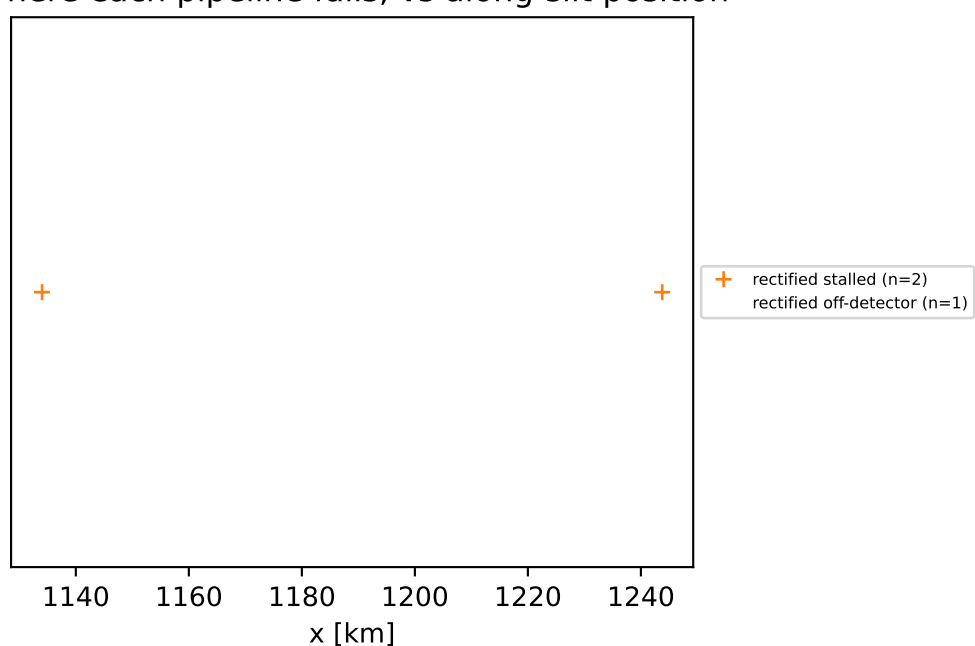
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

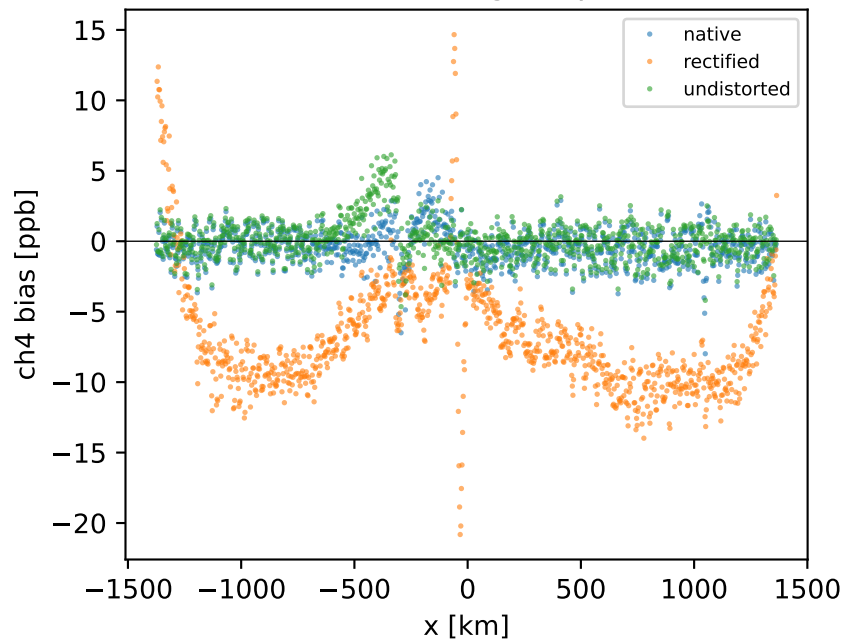


Where each pipeline fails, vs along-slit position

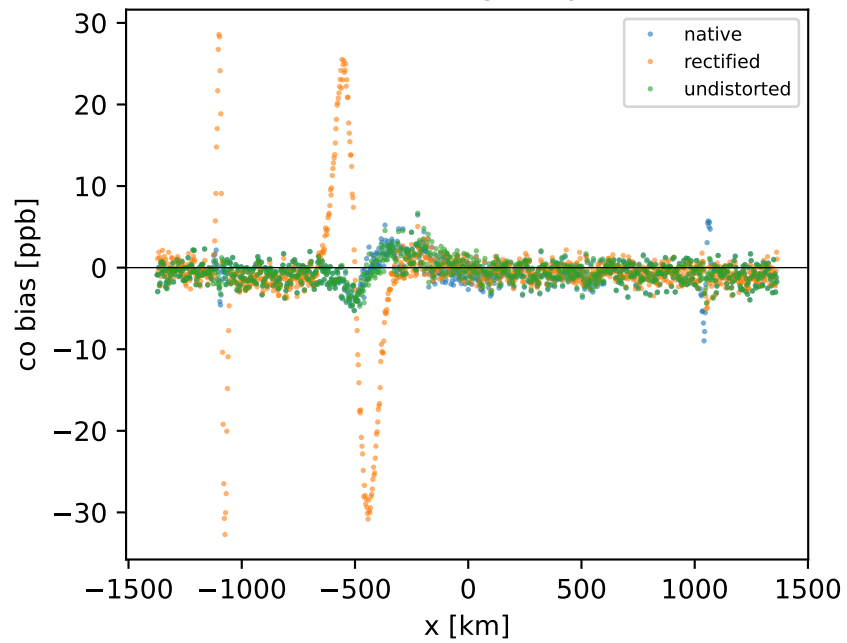


Joint FPA0+FPA3 -- realistic, noise (native/rectified/undistorted)

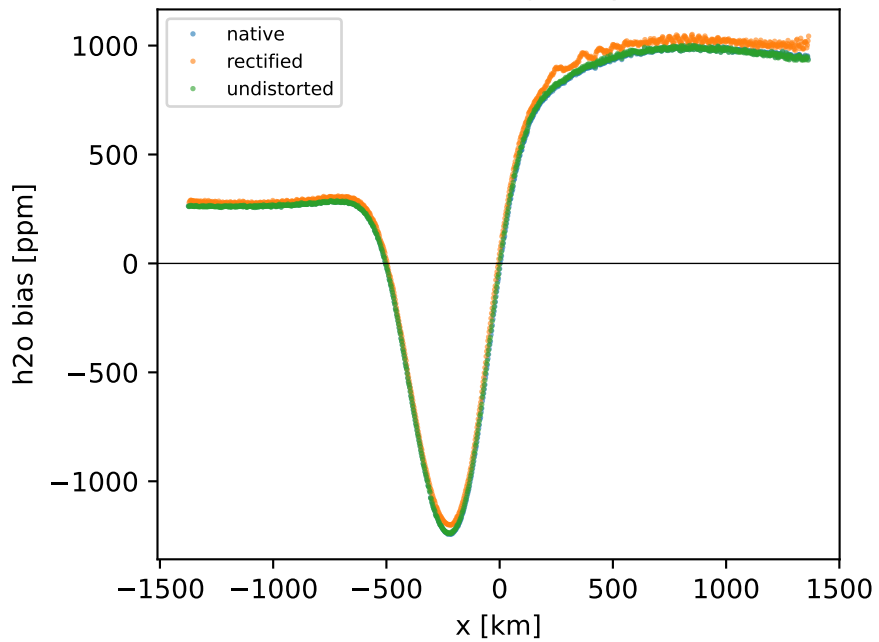
ch4 bias vs along-slit position



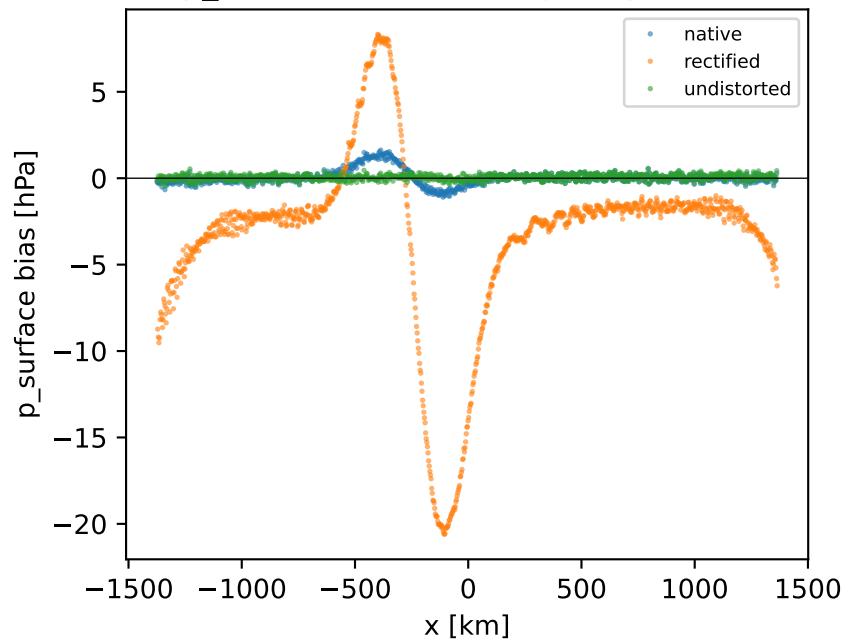
co bias vs along-slit position



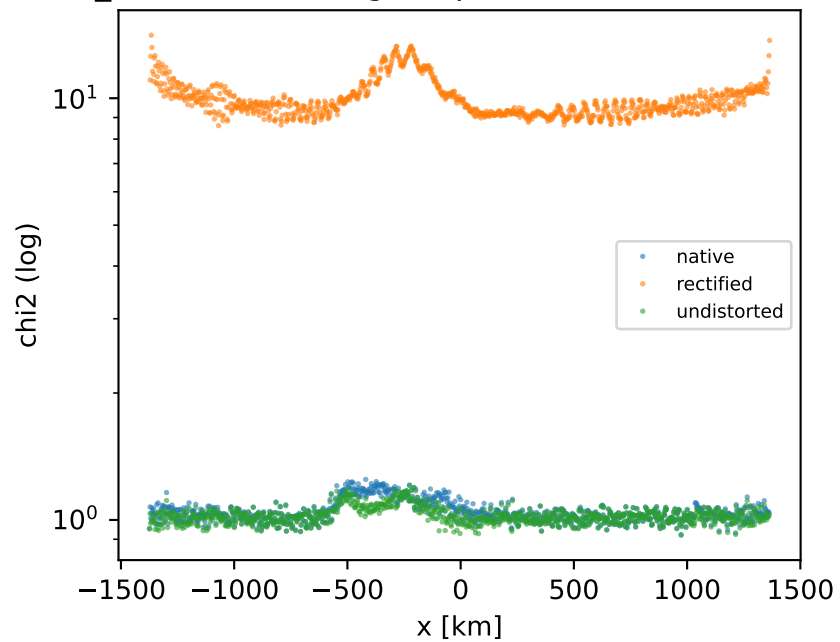
h2o bias vs along-slit position



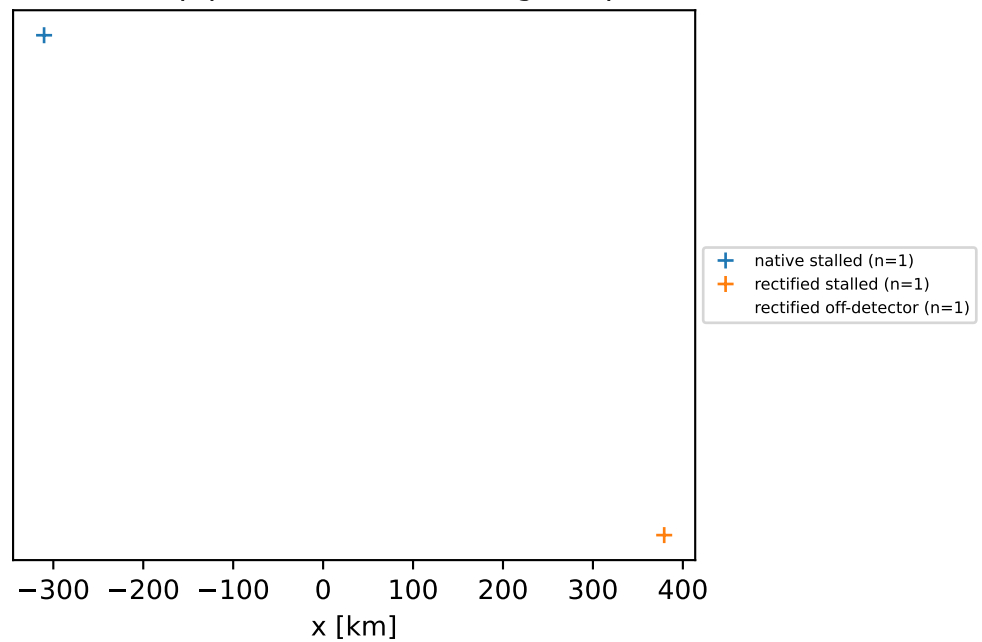
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

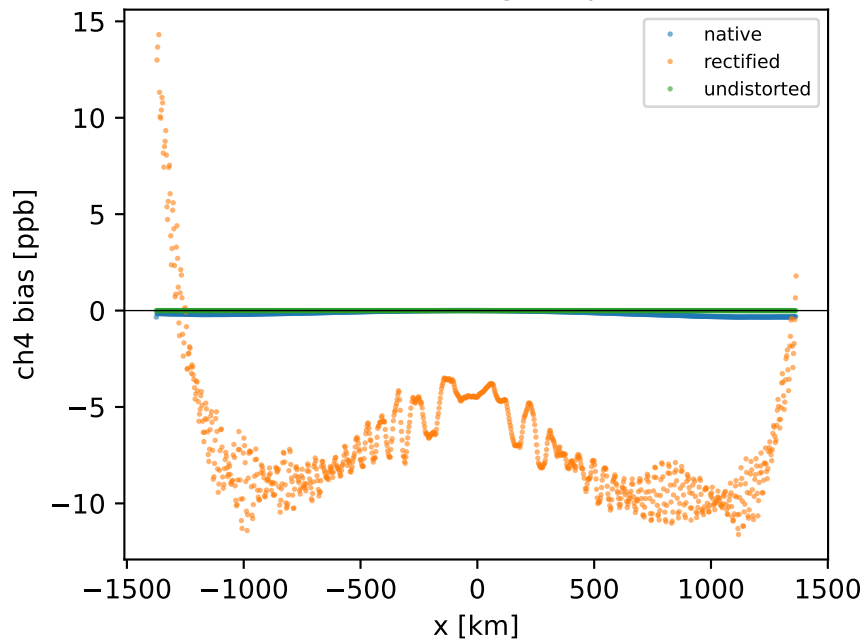


Where each pipeline fails, vs along-slit position

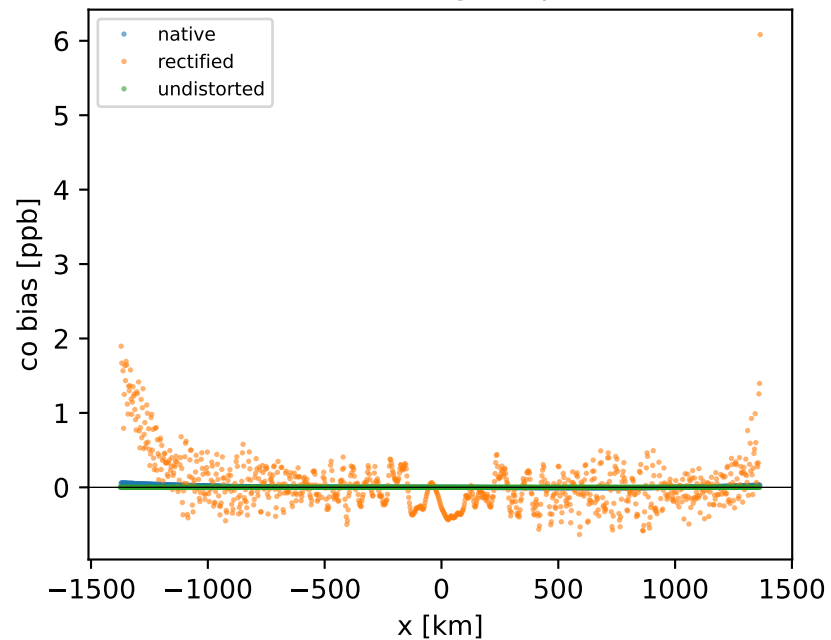


Joint FPA0+FPA3 -- uniform, no noise (native/rectified/undistorted)

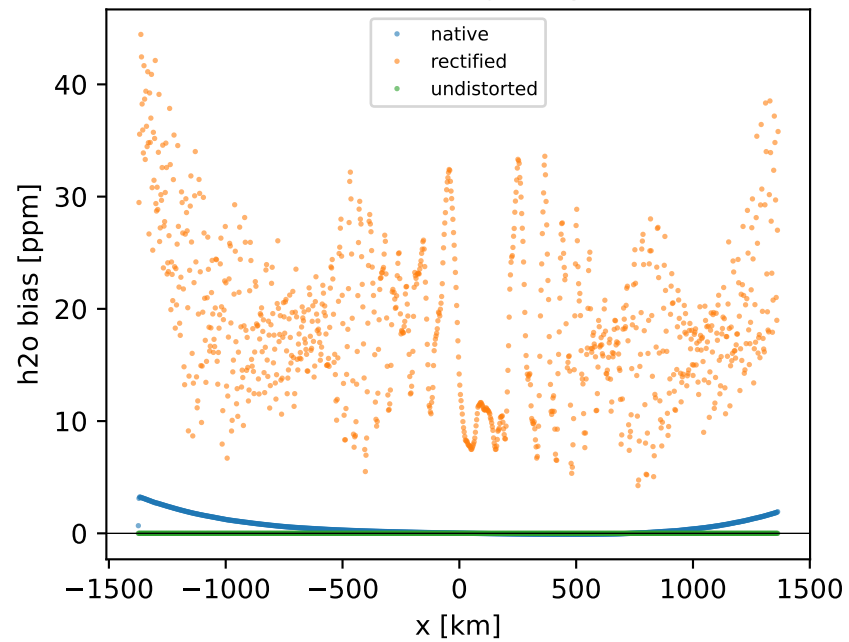
ch4 bias vs along-slit position



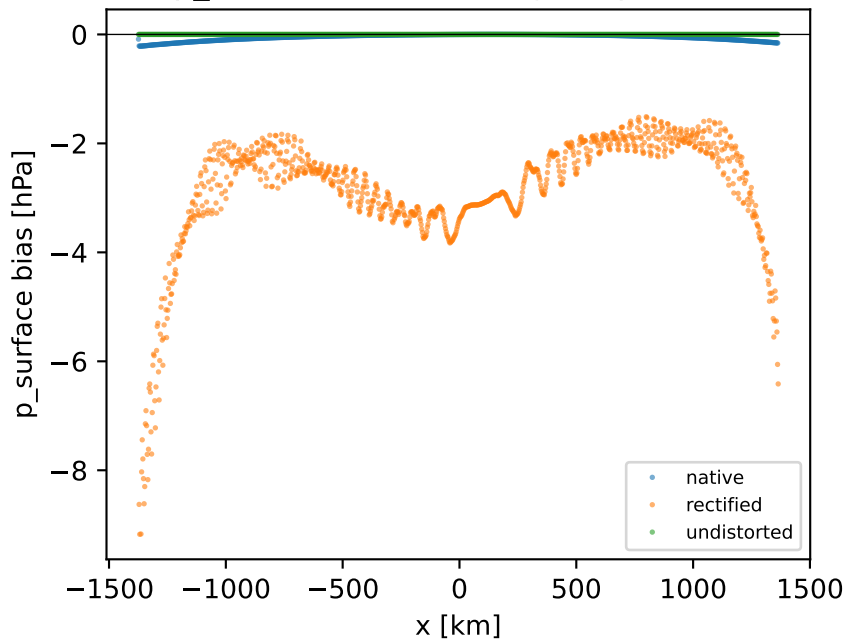
co bias vs along-slit position



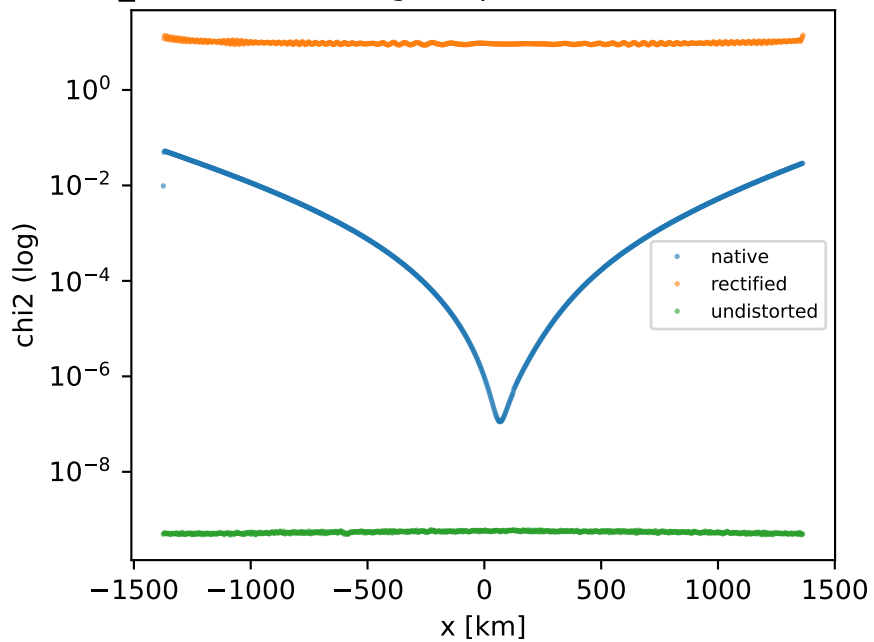
h2o bias vs along-slit position



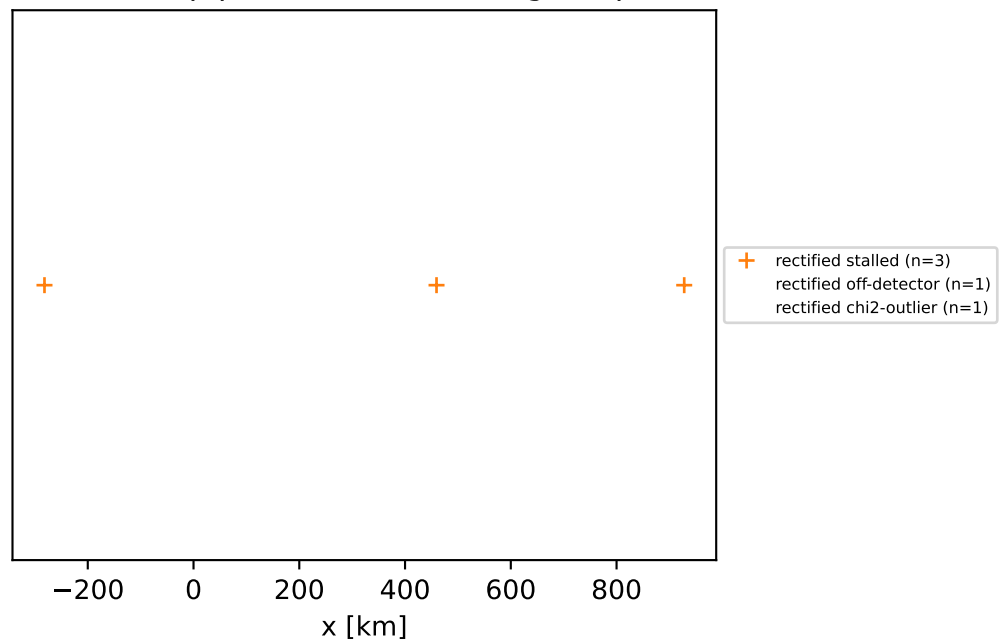
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

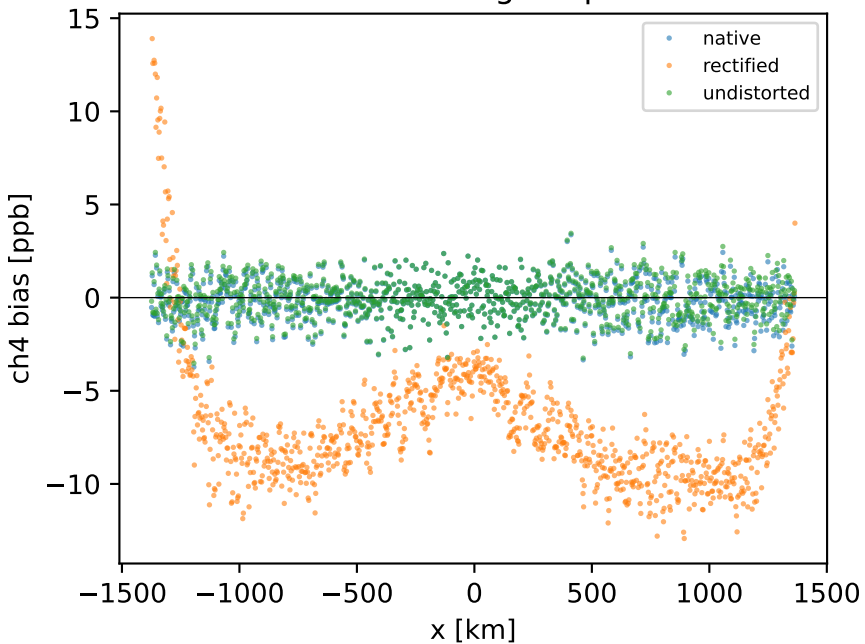


Where each pipeline fails, vs along-slit position

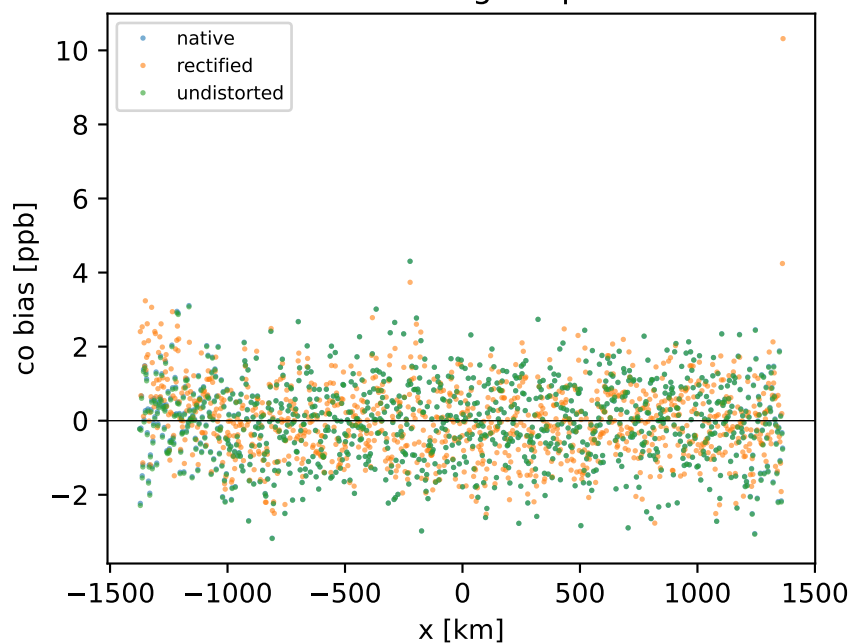


Joint FPA0+FPA3 -- uniform, noise (native/rectified/undistorted)

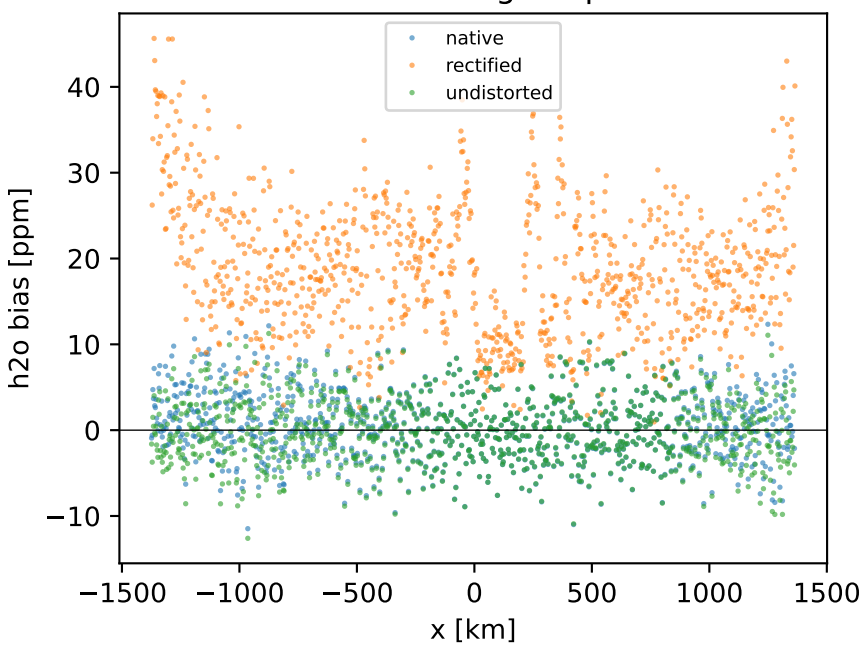
ch4 bias vs along-slit position



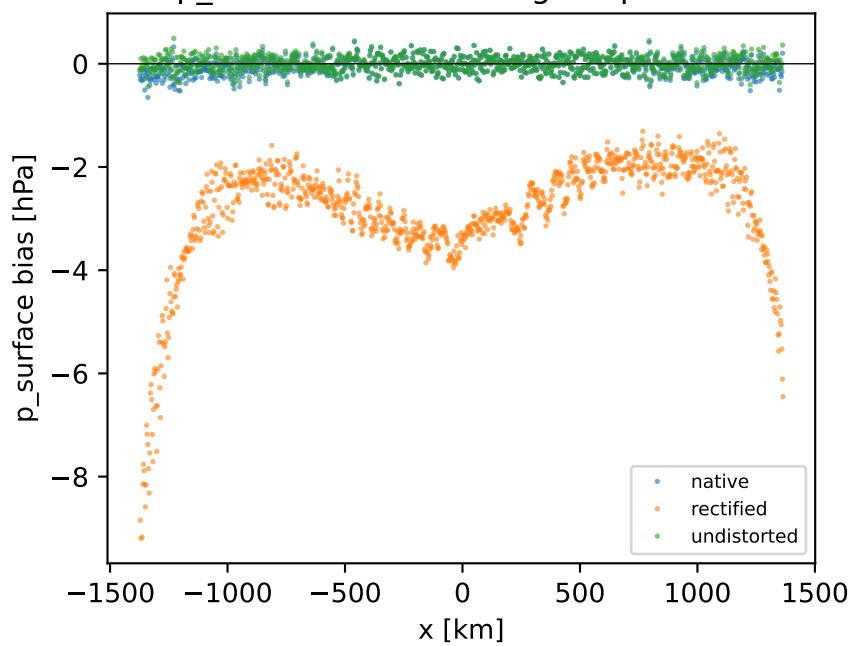
co bias vs along-slit position



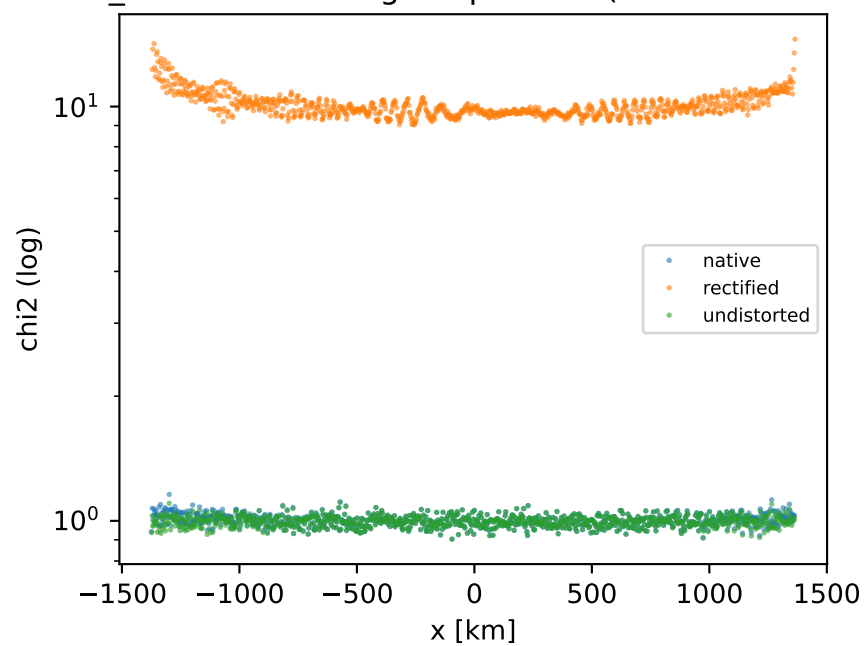
h2o bias vs along-slit position



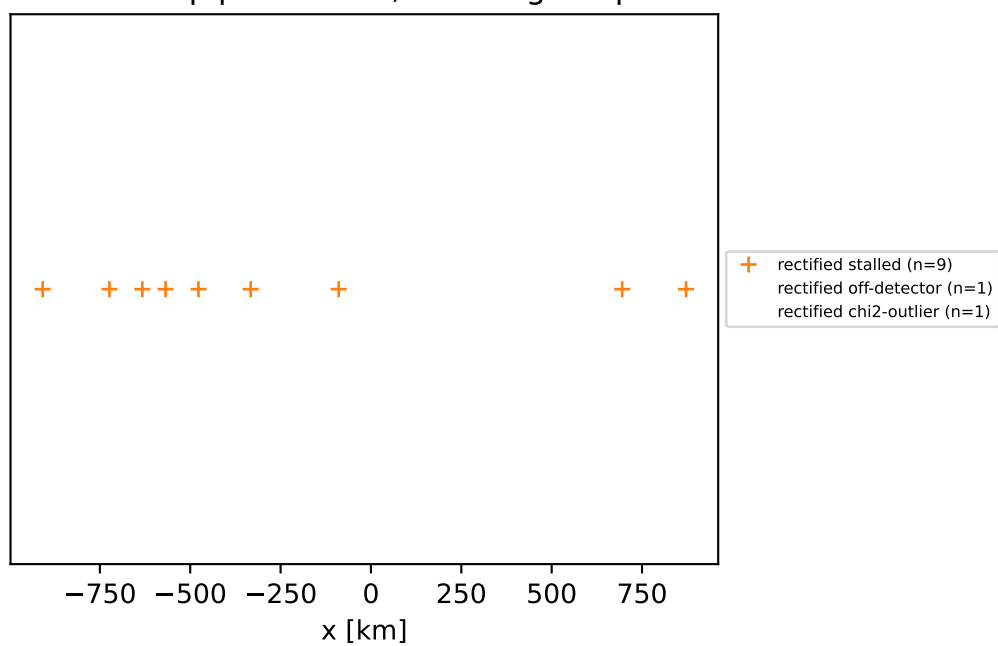
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

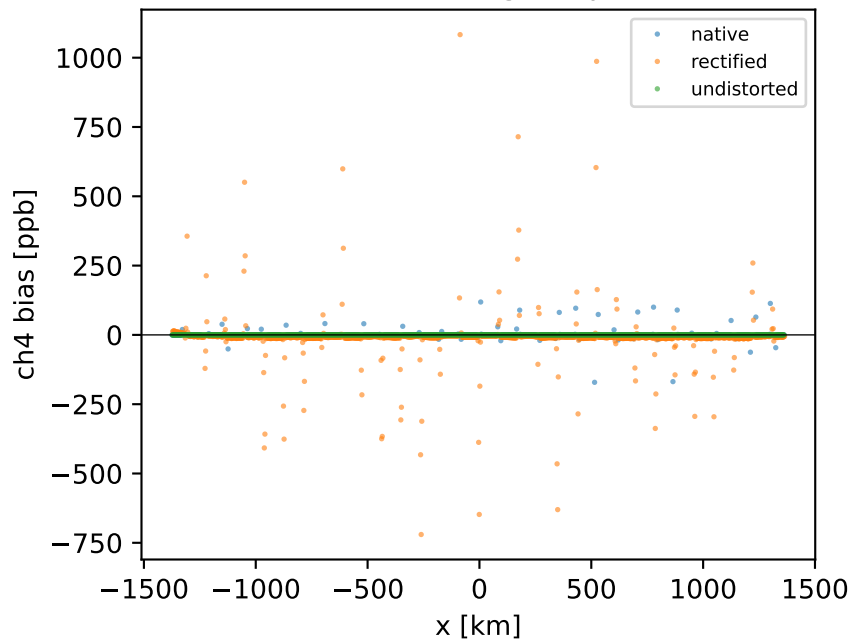


Where each pipeline fails, vs along-slit position

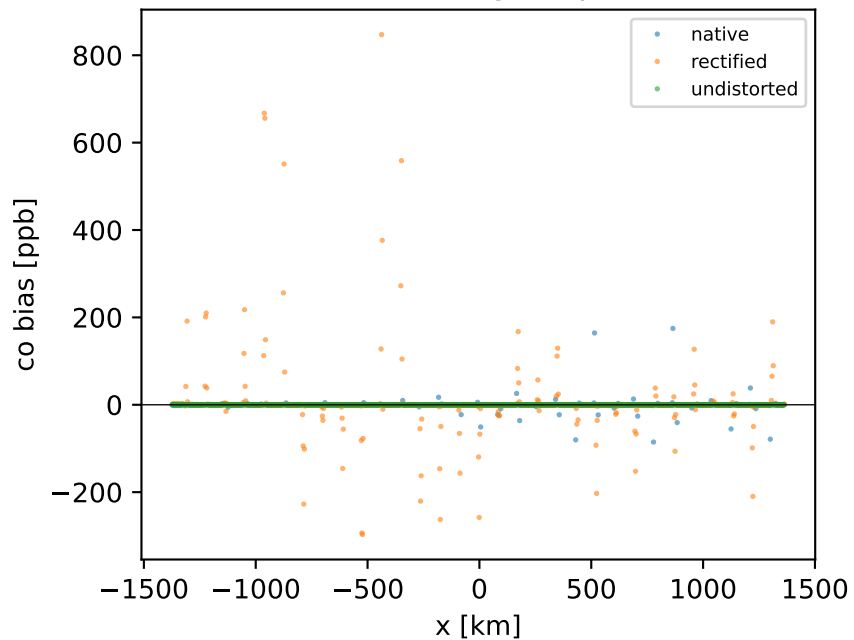


Joint FPA0+FPA3 -- barcode, no noise (native/rectified/undistorted)

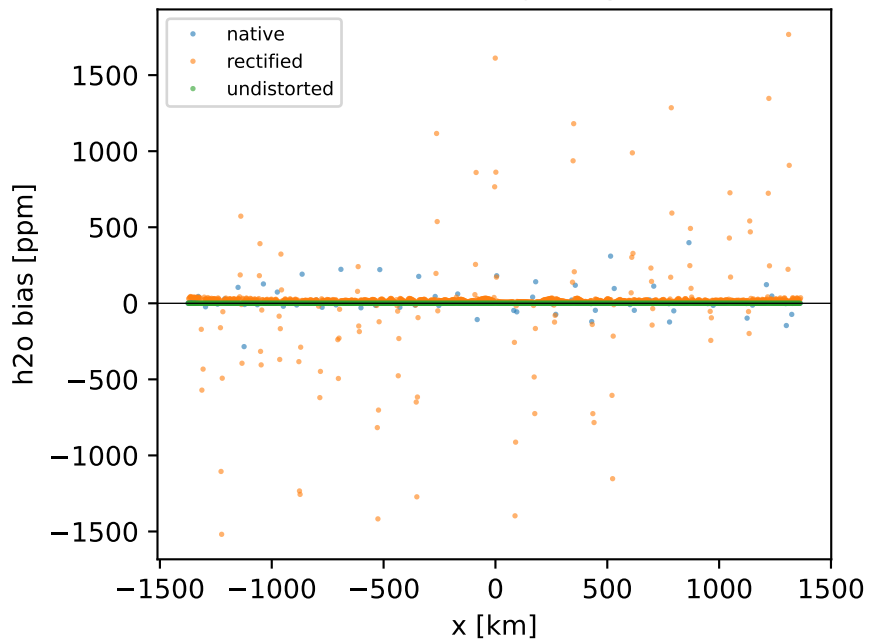
ch4 bias vs along-slit position



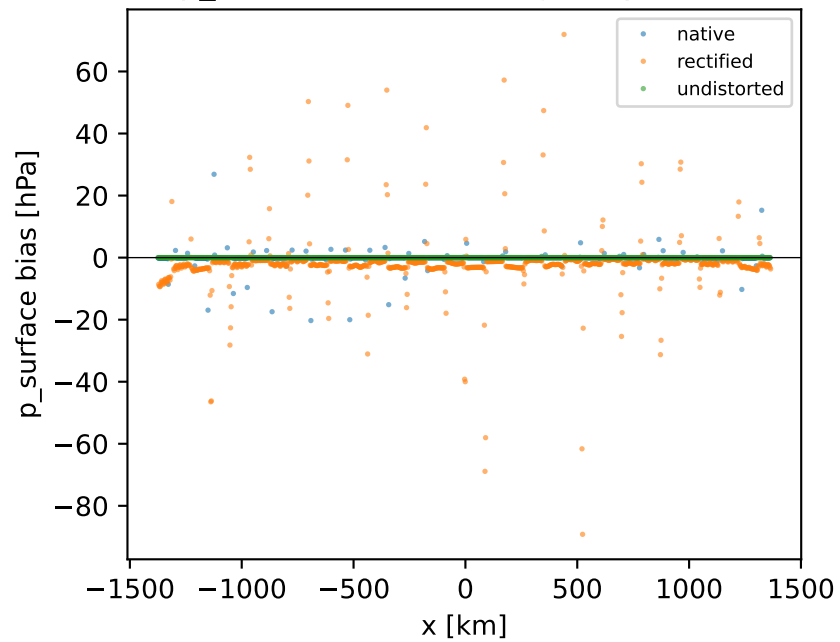
co bias vs along-slit position



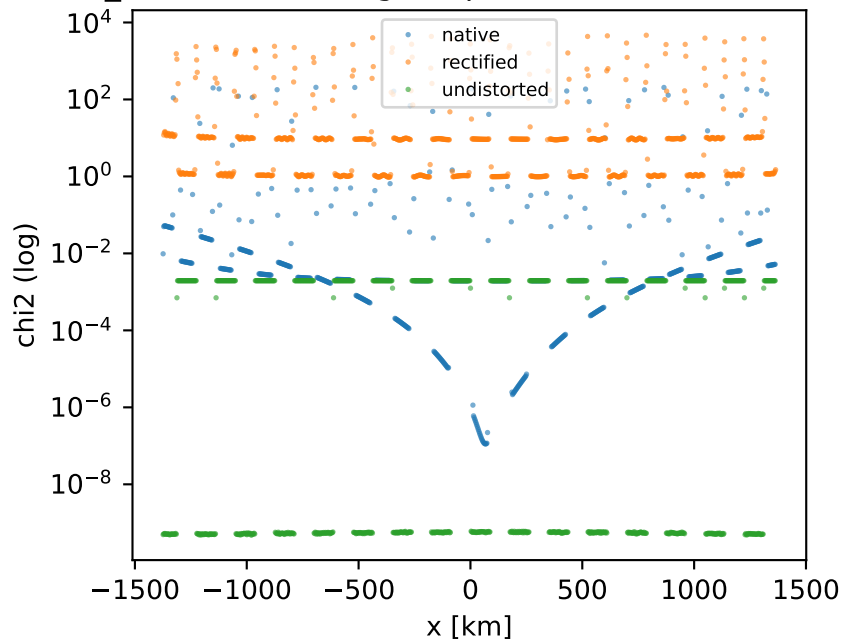
h2o bias vs along-slit position



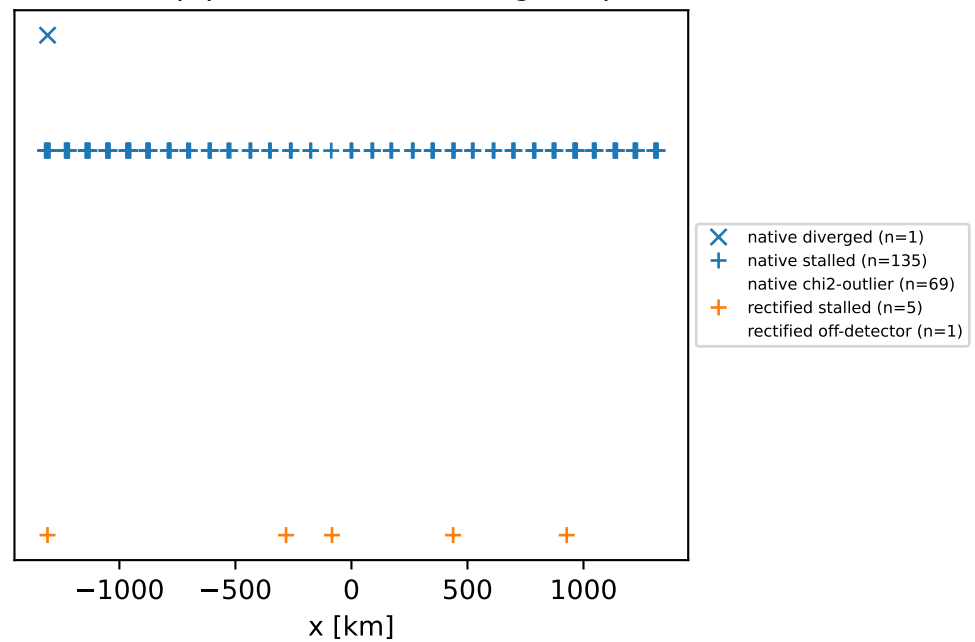
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)

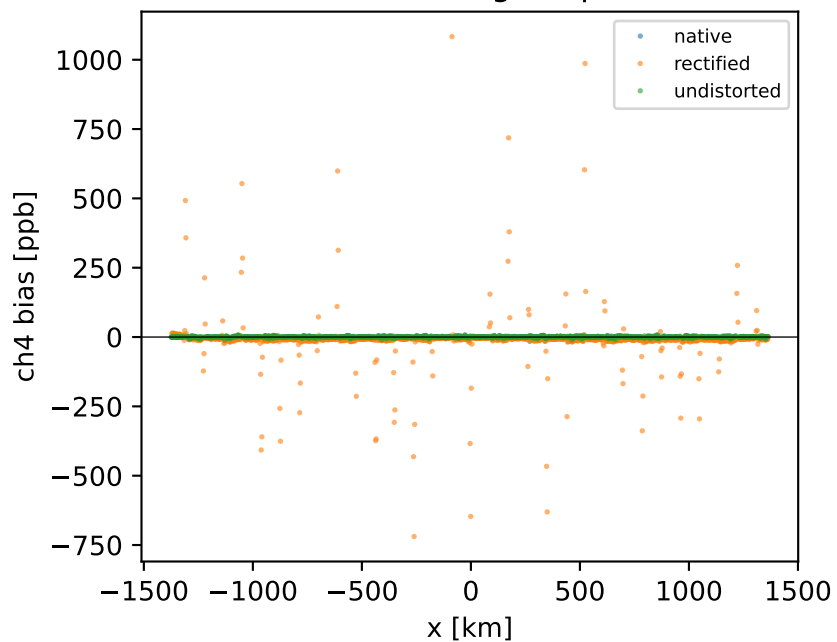


Where each pipeline fails, vs along-slit position

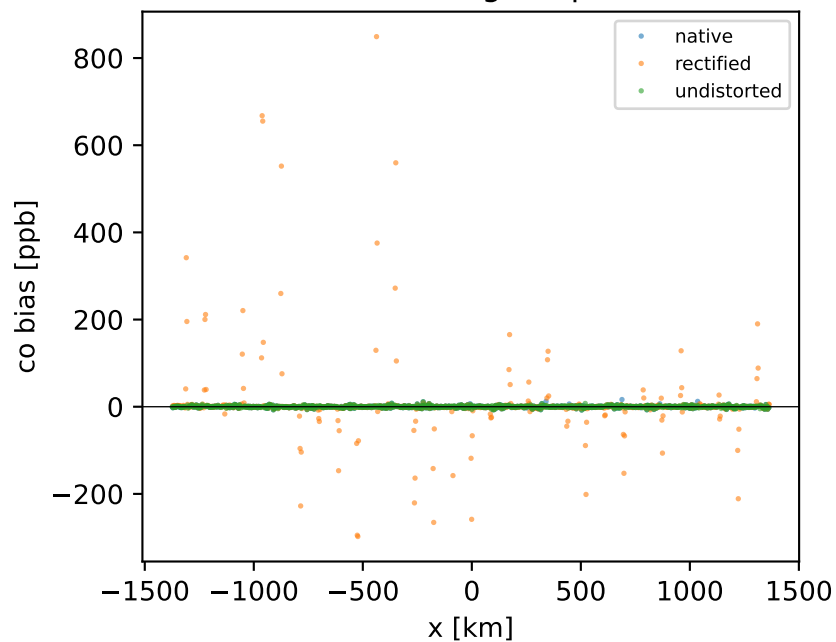


Joint FPA0+FPA3 -- barcode, noise (native/rectified/undistorted)

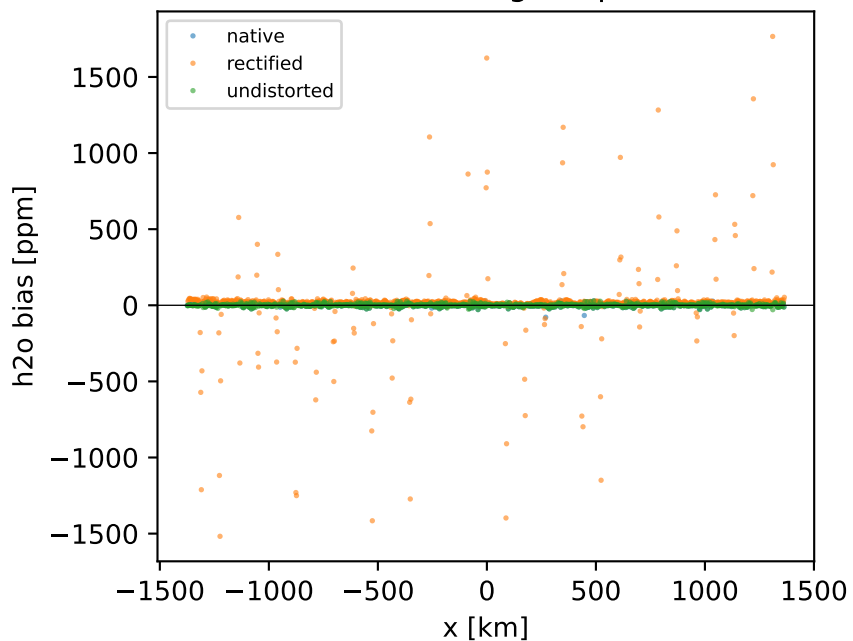
ch4 bias vs along-slit position



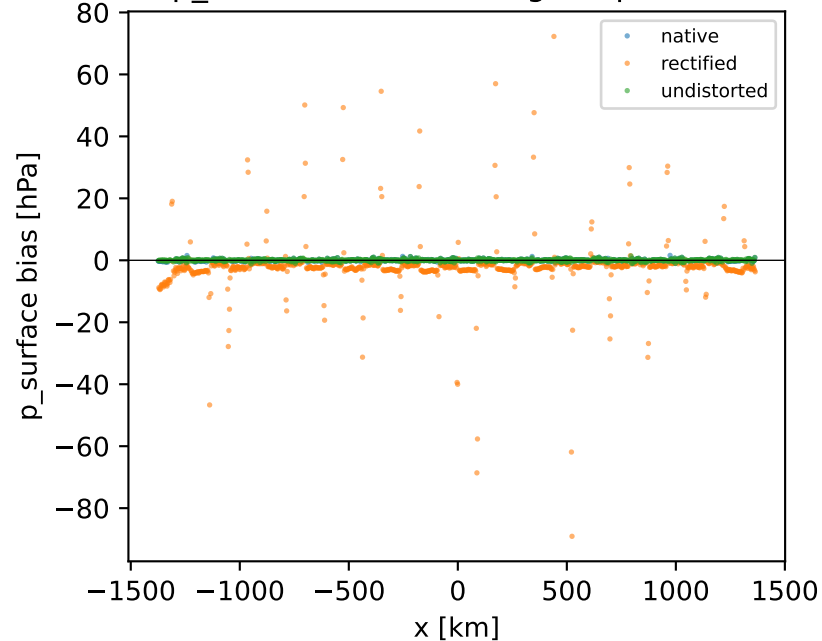
co bias vs along-slit position



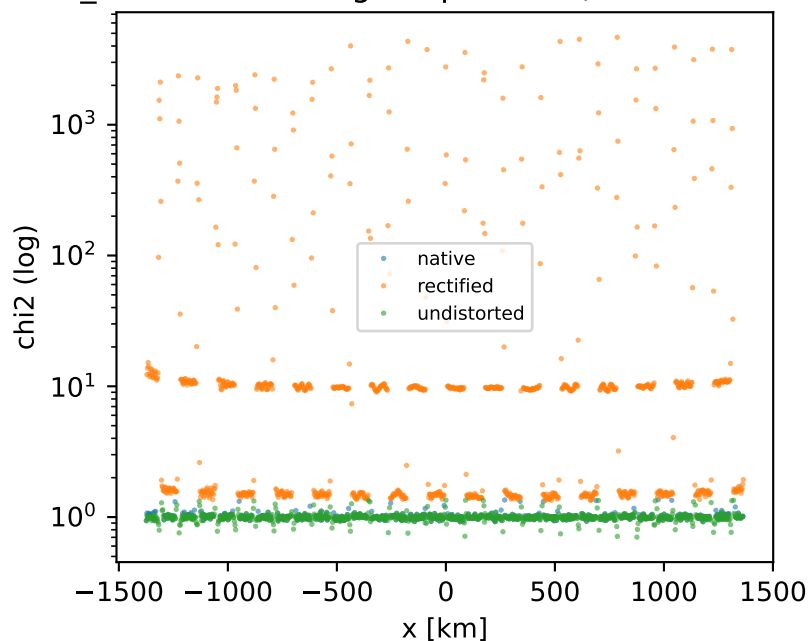
h2o bias vs along-slit position



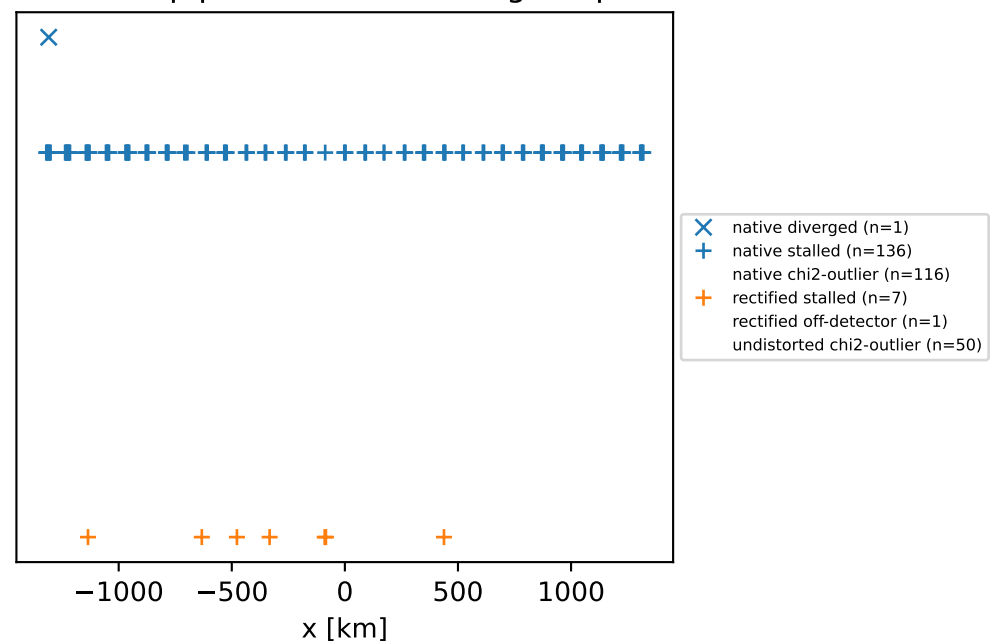
p_surface bias vs along-slit position



chi2_reduced vs along-slit position (outliers excluded)



Where each pipeline fails, vs along-slit position



FPA0+FPA3 -- robust bias statistics across scenes/noise (native/rectified/undistorted) [page 1/2]

scene	noise	quantity	pipeline	n	mean	std	median	MAD-sig	n_out
realistic	0	ch4	native	1010	-0.3413	0.8118	-0.4975	0.2923	42
realistic	0	co	native	1010	-0.5913	1.079	-0.7308	0.1696	157
realistic	0	h2o	native	1010	+379.8	641.9	+283.1	911.6	0
realistic	0	p_surface	native	1010	+0.0431	0.3872	+0.01252	0.09353	111
realistic	0	ch4	rectified	1021	-6.881	4.042	-7.844	3.233	0
realistic	0	co	rectified	1021	-0.7955	6.768	-0.6498	0.6416	117
realistic	0	h2o	rectified	1021	+415.8	645.6	+303.8	951.1	0
realistic	0	p_surface	rectified	1021	-3.382	5.289	-2.239	1.141	120
realistic	0	ch4	undistorted	1010	+0.1107	1.043	-0.2215	0.284	66
realistic	0	co	undistorted	1010	-0.523	0.8552	-0.679	0.1689	154
realistic	0	h2o	undistorted	1010	+382	641	+282.7	918.7	0
realistic	0	p_surface	undistorted	1010	+0.04346	0.03837	+0.05185	0.04317	0
realistic	1	ch4	native	1009	-0.3637	1.33	-0.3447	1.218	0
realistic	1	co	native	1009	-0.6324	1.564	-0.6474	1.261	0
realistic	1	h2o	native	1009	+381.2	640.6	+283.5	912.4	0
realistic	1	p_surface	native	1009	+0.04613	0.4198	+0.008882	0.2335	0
realistic	1	ch4	rectified	1022	-6.888	4.095	-7.786	3.232	0
realistic	1	co	rectified	1022	-0.8284	6.811	-0.7142	1.321	87
realistic	1	h2o	rectified	1022	+416.4	645.5	+304.7	950.7	0
realistic	1	p_surface	rectified	1022	-3.383	5.294	-2.249	1.169	117
realistic	1	ch4	undistorted	1010	+0.08902	1.457	-0.003109	1.205	0
realistic	1	co	undistorted	1010	-0.5603	1.42	-0.588	1.249	0
realistic	1	h2o	undistorted	1010	+382	640.9	+283	921.8	0
realistic	1	p_surface	undistorted	1010	+0.04739	0.1562	+0.04934	0.1612	0
uniform	0	ch4	native	1010	-0.1426	0.1083	-0.1431	0.1301	0
uniform	0	co	native	1010	+0.00721	0.01452	+0.002002	0.008983	0
uniform	0	h2o	native	1010	+0.5723	0.7733	+0.2407	0.4433	0
uniform	0	p_surface	native	1010	-0.05289	0.05492	-0.03233	0.04382	0
uniform	0	ch4	rectified	1019	-6.876	3.492	-7.781	2.423	3
uniform	0	co	rectified	1019	+0.04628	0.3967	-0.004964	0.2748	1
uniform	0	h2o	rectified	1019	+18.79	7.17	+18.05	7.031	0
uniform	0	p_surface	rectified	1019	-2.861	1.098	-2.608	0.8085	2
uniform	0	ch4	undistorted	1010	-0.0002028	1.8e-05	-0.0002016	1.783e-05	0
uniform	0	co	undistorted	1010	-5.251e-06	9.332e-06	-5.379e-06	9.515e-06	0
uniform	0	h2o	undistorted	1010	-0.0002592	0.0001152	-0.0002417	0.0001224	0
uniform	0	p_surface	undistorted	1010	-3.376e-05	6.913e-06	-3.545e-05	5.769e-06	0
uniform	1	ch4	native	1010	-0.1624	1.064	-0.1281	1.059	0
uniform	1	co	native	1010	-0.0319	1.084	+0.004417	1.104	0
uniform	1	h2o	native	1010	+0.5472	3.844	+0.3835	3.926	0
uniform	1	p_surface	native	1010	-0.04886	0.158	-0.04529	0.1551	0
uniform	1	ch4	rectified	1013	-6.875	3.578	-7.6	2.614	1
uniform	1	co	rectified	1013	+0.01752	1.058	-0.009652	0.9771	1
uniform	1	h2o	rectified	1013	+18.79	7.697	+18.16	7.794	0
uniform	1	p_surface	rectified	1013	-2.863	1.11	-2.649	0.8146	2

FPA0+FPA3 -- robust bias statistics across scenes/noise (native/rectified/undistorted) [page 2/2]

uniform	1	ch4	undistorted	1010	-0.0201	1.059	+0.005572	1.047	0
uniform	1	co	undistorted	1010	-0.03911	1.084	+0.003805	1.108	0
uniform	1	h2o	undistorted	1010	-0.02508	3.79	-0.1923	3.804	0
uniform	1	p_surface	undistorted	1010	+0.004032	0.148	+0.004795	0.1509	0
barcode	0	ch4	native	805	+0.5656	14.43	-0.1341	0.1758	75
barcode	0	co	native	805	-0.1046	10.71	+0.001273	0.008236	85
barcode	0	h2o	native	805	+2.411	30.53	+0.2175	0.4017	69
barcode	0	p_surface	native	805	-0.1062	2.023	-0.01385	0.02022	86
barcode	0	ch4	rectified	1018	-10.6	90.89	-9.289	3.717	93
barcode	0	co	rectified	1018	+2.586	59.58	-0.07796	0.317	121
barcode	0	h2o	rectified	1018	+13.58	204.8	+18.4	8.746	97
barcode	0	p_surface	rectified	1018	-1.905	8.439	-1.955	1.492	52
barcode	0	ch4	undistorted	1010	+1.495e-05	0.000222	-0.0001619	0.0002145	0
barcode	0	co	undistorted	1010	+8.238e-05	8.876e-05	+1.838e-05	7.306e-05	0
barcode	0	h2o	undistorted	1010	-8.011e-05	0.0002241	-8.901e-05	0.0002473	0
barcode	0	p_surface	undistorted	1010	-1.514e-05	2.034e-05	-1.209e-05	2.996e-05	0
barcode	1	ch4	native	757	-0.3259	2.728	-0.1458	1.784	3
barcode	1	co	native	757	-0.003651	2.333	-0.06857	1.781	1
barcode	1	h2o	native	757	+0.2116	9.538	+0.1932	6.105	2
barcode	1	p_surface	native	757	-0.004671	0.3378	-0.02361	0.2426	0
barcode	1	ch4	rectified	1016	-10.27	92.31	-8.873	4.049	89
barcode	1	co	rectified	1016	+2.985	60.67	-0.1202	1.714	87
barcode	1	h2o	rectified	1016	+12.17	208.6	+18.53	10.69	92
barcode	1	p_surface	rectified	1016	-1.826	8.366	-1.99	1.497	52
barcode	1	ch4	undistorted	960	-0.02291	2.288	+0.02237	1.706	0
barcode	1	co	undistorted	960	-0.01502	2.103	-0.01647	1.684	0
barcode	1	h2o	undistorted	960	-0.1027	8.469	-0.3217	5.78	0
barcode	1	p_surface	undistorted	960	+0.01918	0.3115	+0.007623	0.2234	0